Field Hours

CSCE 632 Accessible Computing

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Where the barrier sits

You need a model before you choose a fix, because the model decides where the problem sits. A medical model puts the problem in a body and treats a design change as an exception. An interaction model puts the problem in a mismatch between a person and an environment built for someone else. A door with 12 stairs is a barrier for a wheelchair user and for a person with a cart. You change the entrance.

The model changes the requirement you write. A form that works only with a mouse fails the interaction model even if all 8 people in your last study used a mouse. Those 8 matched the old assumption, so they could not reveal the barrier. A keyboard path is part of the task. You also do not demand a disability disclosure before the page works, and you do not collect that disclosure if 0 features need it.

Two places to put the problem.

Medical model

The body is the problem

The design is an exception

Interaction model

The mismatch is the problem

You change the 12 stairs

Twelve stairs and a cart

1. The entrance is 12 stairs and no ramp.

2. A wheelchair user and a person with a cart both stop. The stairs are the barrier in the interaction model.

3. Adding a ramp changes the entrance. Asking those people to identify themselves before they may enter does not.

Problem 1. Your last study had 8 mouse users and the form works only with a mouse. Why does that study not clear the form?

Answer. Those 8 matched the assumption that everyone uses a mouse, so they could not reveal the barrier. Under the interaction model the keyboard path is part of the task, not an exception you defer.

Problem 2. A door has 12 stairs. Where does the interaction model locate the problem, and what do you change?

Answer. It locates the problem in the mismatch between the person and an entrance built for someone else. You change the entrance. The same 12 stairs also stop a person with a cart.

Lecture 2 · 90 minutes

A statute points at a standard

Law and a technical standard are different layers you have to keep straight. The US ADA is a civil-rights statute, and Section 508 is a federal procurement rule. The 2017 refresh of Section 508 points federal web content at WCAG 2.0 Level AA, a list of 38 success criteria. A private shop and a federal site are not under the same rule. You name the rule that applies before you call a page covered.

WCAG itself is not a statute. It becomes an obligation when a law, a contract, or a policy incorporates it. Level A is the floor, and Level AA adds criteria on top of A. You can fail AA while passing A. Contrast of body text is an AA criterion in WCAG 2.0, set at 4.5 to 1, and a page can meet 20 other criteria and still fail that one. Counting passed criteria does not average a failure away.

From a statute down to one check.

Name the statute or the contract

See which WCAG level it names

Level AA adds criteria past A

One failed criterion fails the level

Twenty passes do not erase one fail

1. The policy incorporates WCAG 2.0 Level AA.

2. Body text on the page has contrast 2 to 1. The AA line for ordinary text is 4.5 to 1.

3. You can pass 20 other criteria and still fail AA. The failure is not averaged away.

Problem 1. What does the 2017 refresh of Section 508 point federal web content at, and how many success criteria is that level in WCAG 2.0?

Answer. WCAG 2.0 Level AA, a list of 38 success criteria. Section 508 is a federal procurement rule. A private shop is not under that same rule.

Problem 2. A page meets 20 AA criteria and fails body-text contrast at 2 to 1. Have you met Level AA?

Answer. No. Ordinary text needs 4.5 to 1 under WCAG 2.0 Level AA. You can fail AA while passing A, and counting passed criteria does not average one applicable failure away.

Lecture 3 · 90 minutes

One design, more people

Universal design aims at one design that more people can use, rather than a separate door for a labeled group. A curb cut is the usual evidence: it was pressed for wheelchair users and it also helps a traveler with a suitcase. You are looking for that kind of overlap on purpose. Flexibility is one of the operating principles. A site with exactly

1 path through a form, and no way to skip a repeated block, fails flexibility.

Perceptible information and tolerance for error are two further principles. You do not convey status by color alone, because someone who cannot separate the colors loses the status. Pair the color with a word. Tolerance for error means a destructive action can be undone or must be confirmed. Deleting 30 files with no undo fails that test. A confirm step, or an undo of a few seconds, is the mechanism you add.

A principle you can fail.

1. Ask who else the change helps

2. Offer more than 1 path

3. Do not use color alone

4. Confirm before deleting 30 files

Color is not the status

1. A form shows errors only by turning the border red.

2. A reader who cannot separate red from the default border loses the status. You add the word 'error' beside the field.

3. The delete control removes 30 files immediately. You add a confirm, because tolerance for error requires undo or confirmation.

Problem 1. A form has exactly 1 path and no way to skip a repeated block. Which principle fails, and what would a curb-cut style change look like?

Answer. Flexibility fails. Universal design looks for one design that more people can use. A second way through, including a skip of the repeated block, is the overlap you are looking for, the way a curb cut helps more than one group.

Problem 2. Delete removes 30 files with no undo, and errors are shown only in red. Which two principles fail?

Answer. Tolerance for error fails on the delete, because a destructive action needs undo or a confirm. Perceptible information fails on the red border, because status by color alone is lost when the colors cannot be separated.

Lecture 4 · 90 minutes

Four groups of criteria

WCAG groups criteria so you can see what kind of barrier you are looking for. Perceivable means the information is available to at least 1 sense the person can use. A photograph with no text alternative fails this group, because a screen reader has nothing to speak. Operable means the controls can be used. A widget that responds only to a hover and ignores the keyboard fails operable. A captioned video can still trap keyboard focus.

Understandable means text and behavior are predictable enough to use. A form that moves focus on the first keystroke fails that group. The fourth group asks that content keep working with user agents, including assistive

technologies. You meet it with markup those agents already understand, not with a picture of text. A custom control needs a name, a role, and a value. If you ship 1 div that only looks like a button, this is the group you failed.

What kind of barrier is this?

Perceivable: a sense can reach it

Operable: a control can be used

Understandable: behavior is predictable

Markup agents already understand

A caption does not free the keyboard

1. The video has captions, so the speech is perceivable as text.

2. The player moves focus in and will not release it. The keyboard user cannot reach the rest of the page.

3. That failure is operable, not perceivable. You do not close the ticket because the captions exist.

Problem 1. A photograph has no text alternative. Which group fails, and why does a screen reader care?

Answer. Perceivable. The screen reader has nothing to speak, so the information is not available to a sense the person can use.

Problem 2. You ship 1 div that only looks like a button. Which group fails, and what three things does a custom control need?

Answer. The group about working with user agents and assistive technologies. A custom control needs a name, a role, and a value. Markup those agents already understand is the mechanism, not a picture of text.

Lecture 5 · 90 minutes

The accessibility tree

A screen reader walks an accessibility tree, not your pixels. Each object needs a name a person hears, a role such as button, and a state such as checked. A native button exposes all 3. A div with a click handler and no role is easy to miss in that tree. The visible label is not automatically the name. You bind the visible text to the control, or the spoken name and the label drift apart.

Keyboard use is the other half of operable. Tab moves through controls in the order you set, and the focused control has to be visible. A focused button should activate from Enter and from Space, which is 2 keys, not a mouse click. If the div listens only to click, both keys do nothing. Keep a visible focus indicator. Run 1 complete task with the mouse unplugged before you call the path done.

From pixels to a spoken control.

Expose the accessibility tree -> Give a name, role, and state -> Bind the visible label -> Accept Enter

and Space

A div that only looks like Search

1. The control is a div. The visible word is Search. There is no role and no accessible name.

2. A screen reader walking the tree does not announce a button named Search.

3. You set the role to button, bind the name to the visible word, and handle Enter and Space. Tab order puts it after the field, and the focus ring stays visible.

Problem 1. The visible label says Search, but the accessible name is empty. Why can those two differ, and what do you bind?

Answer. The visible label is not automatically the name in the accessibility tree. You bind the visible text to the control. Otherwise the spoken name and the label drift apart.

Problem 2. A focused control listens only to click. What do Enter and Space do, and what test do you run before calling the path done?

Answer. Both keys do nothing. A button should activate from those 2 keys. You run 1 complete task with the mouse unplugged, and you keep a visible focus indicator.

Lecture 6 · 90 minutes

Contrast, captions, and hit targets

Ordinary text under WCAG 2.0 Level AA needs contrast of at least 4.5 to 1 against its background. Large text, about 18 point regular or 14 point bold, may go down to 3 to 1. You measure the pixels on the screen, not the brand swatch from before it was lightened. Gray at 2 to 1 on white fails even when the page looks quiet. Darken the text or lighten the card. A help sentence does not raise the ratio.

Captions are timed text locked to the audio. A transcript under the player is a different aid. A viewer of a 4 minute clip needs the line when it is spoken, which a transcript does not time. Later WCAG also sets a pointer target at 24 by 24 CSS pixels, with some exceptions. You miss that if the visible icon is 24 pixels and the hit region is still 10. Measure the hit region, not the picture of the icon.

Two different repairs.

Contrast

Ordinary text needs 4.5 to 1

Measure the pixels

A help note does not count

Target size

Target at least 24 by 24

The hit region can be 10

Measure the hit, not the icon

Gray type on a white card

1. You sample the text pixel and the card pixel. The contrast is 2 to 1.

2. Ordinary text needs 4.5 to 1. This fails, even if a designer calls the gray quiet.

3. You darken the text until the measured ratio meets 4.5 to 1. You do not add a sentence to the help page and leave the gray.

Problem 1. Body text measures 2 to 1 on white. Large text is about 18 point. Which threshold applies to the body text, and does a help sentence fix it?

Answer. Body text needs 4.5 to 1 under WCAG 2.0 Level AA. The 3 to 1 allowance is for large text, about 18 point regular or 14 point bold. A help sentence does not raise the ratio.

Problem 2. The icon art is 24 pixels wide and the hit region is 10 pixels. A 4 minute clip has a transcript but no captions. What fails?

Answer. The hit target fails if the requirement is 24 by 24 CSS pixels, because you measure the hit region, not the picture. Captions also fail: a transcript is not timed text, and a viewer of the 4 minute clip needs the line when it is spoken.

Lecture 7 · 90 minutes

An audit and ten papers

An automated scan is a filter, not an audit. It can catch a missing text alternative and a contrast ratio. It cannot see that tab order jumps from the header to the footer and skips the form. Run the scanner, then do 1 task by keyboard and 1 task with a screen reader. A page can show 6 machine-reported failures and still hide 3 focus-order bugs you only find by hand. Record 9.

The literature half is yours to write. Pick 10 papers on one question, such as how screen-reader users complete a checkout, and build a matrix. Rows are papers. Columns are the population, the task, the method, and the claim, which is 4 columns. A paragraph that only praises the topic has not used the 10. You want the disagreement: two studies of the same task with different error counts.

A scan, then two manual passes.

Run the automated scan -> Finish 1 keyboard task -> Finish 1 screen-reader task -> Add the bugs the scan missed

Six from the tool, three from you

1. The scanner reports 6 contrast failures and 0 keyboard failures.

2. Your keyboard pass finds 3 focus-order bugs. The screen-reader pass confirms the search control has no name.

3. You record 9, not 6. The literature matrix is a separate product: 10 papers, 4 columns, one question.

Problem 1. The scanner reports 6 failures and your keyboard pass finds 3 more. What total do you record, and why is 6 not the audit?

Answer. You record 9. An automated scan is a filter. It misses focus-order bugs, so its 6 was never the full set.

Problem 2. What belongs in the matrix for 10 papers, and why is a paragraph of praise not a review?

Answer. Rows are the 10 papers. Columns are population, task, method, and claim, which is 4 columns. Praise does not use those cells. You are looking for disagreement, such as two studies of one task with different error counts.

Lecture 8 · 90 minutes

A criterion plus an observation

You can now name the model a design assumes, the rule that pulls in a standard, and the criterion a page fails. You can check contrast, walk a keyboard path, and require a name, a role, and a state on a control. You can reduce 10 papers to a matrix of population, task, method, and claim. A finding names a criterion and an observation. A nameless div used as search is a finding, because the tree has nothing to announce.

Keep the legal layer and the tree on separate lines. Under WCAG 2.0 Level AA, body text at 2 to 1 contrast fails even when the button name is correct. Contrast at 7 to 1 does not repair a button with no accessible name. You file those as 2 failures because a team will fix one and close the ticket. The paper matrix can tell you whether the name should match the visible word.

What a finding has to contain.

The model you are using

The rule that applies

The criterion that failed

The observation, not a mood

Two failures, one ticket

1. Body text measures 2 to 1. The search control is a div with no accessible name.

2. Under WCAG 2.0 Level AA those are 2 failures. Contrast at 7 to 1 would fix only the first.

3. You file both. The matrix of 10 papers may tell you the name should be the visible word Search, and the audit says this page does not expose that name.

Problem 1. Body text is 2 to 1 and the search control has a correct accessible name. Under WCAG 2.0 Level AA, is the page done?

Answer. No. Body text fails the 4.5 to 1 contrast criterion even when the name is correct. A correct name does not repair contrast, and you do not average the failure away.

Problem 2. Contrast is 7 to 1 and the search control is a div with no name. Combine the standard and the tree: how many failures do you file, and why will one fix not close both?

Answer. You file 2 failures. Contrast at 7 to 1 can satisfy the contrast criterion and still leave the tree with nothing to announce. A team that fixes only contrast will close the ticket too early. The name is a

separate criterion.